Physical Observer Geometry, Protocol Holonomy, Order by Non-Closure, and Spectral Obstructions

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Abstract

A categorical agency base, a probabilistic-semantic opfibred assignment, a physical observer geometry, and a protocol category are defined. Exact coincidence of semantic profiles implies observer identity relative to separating query families. Local scientific agreement is controlled by an agreement modulus and an agreement radius. Under local continuity and a positive flattenability threshold, the agreement radius is strictly positive and finite. Abstract protocol holonomy is defined functorially; independently, realized path residue and a ball-framed path modulus are introduced, the latter yielding quantitative upper bounds on agreement radii, while local semantic dilatation yields lower bounds. Finite-bank estimators converge under a transfer modulus and an isolated threshold-crossing condition. An explicit binary system–instrument–environment model is solved in closed form, exhibiting exact semantic rigidity, nontrivial commutator residue, and finite agreement windows. Finally, a conditional Yang–Mills reduction is formulated: exponential clustering of gauge-invariant Euclidean correlators implies a spectral mass gap, while the unresolved content is isolated into two bridge statements linking persistent UV/IR abstract holonomy and infrared curvature to exponential clustering.

1 Conventions

Throughout:

1. $\mathcal{A}$ denotes a small category.
2. For a pseudometric space (X, d), the open ball convention is

$$B_r(x) := \{y \in X : \ d(x, y) < r\}.$$

3. If a supremum is taken over the empty subset of $[0, \infty)$, its value is declared to be 0.
4. For a measurable space X, $\mathcal{P}(X)$ denotes the set of probability measures on X.

2 Agency base and probabilistic-semantic opfibred assignment

Definition 2.1 (Agency base)**.** An *agency base* is a small category $\mathcal{A}$. Its objects are neighborhoods of agency. A morphism $f : a \to b$ is an admissible transport of local access structure from a to b.

Definition 2.2 (Local probabilistic-semantic chart)**.** For $a \in \mathrm{Ob}(\mathcal{A})$, a local chart is a tuple

$$U_a = (X_a, \Sigma_a, \lambda_a, \mathcal{G}_a, \tau_a, \mathcal{Q}_a),$$

where:

1. X_a is a standard Borel space;
2. $(X_a, \Sigma_a, \lambda_a)$ is a reference measure space;
3. $\mathcal{G}_a \subseteq \Sigma_a$ is an accessible σ-algebra;
4. $\tau_a : X_a \to S$ is a measurable semantic map into a semantic target space S;
5. $\mathcal{Q}_a \subseteq L^\infty(S)$ is a family of admissible semantic queries.

Definition 2.3 (Density fibre)**.** For $a \in \mathrm{Ob}(\mathcal{A})$, define

$$\mathcal{D}_a := \left\{ \rho \in L^1(X_a, \lambda_a) : \rho \geq 0,\ \int_{X_a} \rho \, d\lambda_a = 1 \right\}.$$

For $\rho \in \mathcal{D}_a$, define

$$\mu_{a,\rho}(E) := \int_E \rho \, d\lambda_a, \qquad E \in \Sigma_a.$$

Definition 2.4 (Semantic law and evaluation)**.** For $\rho \in \mathcal{D}_a$, define

$$\nu_{a,\rho} := (\tau_a)_\#(\mu_{a,\rho}) \in \mathcal{P}(S).$$

For $q \in \mathcal{Q}_a$, define

$$P_{a,\rho}(q) := \int_S q(s) \, d\nu_{a,\rho}(s) = \int_{X_a} q(\tau_a(x)) \, \rho(x) \, d\lambda_a(x).$$

Proposition 2.5 (Uniqueness under full event equality)**.** *Let $\rho, \eta \in \mathcal{D}_a$. If*

$$\mu_{a,\rho}(E) = \mu_{a,\eta}(E) \qquad \forall E \in \Sigma_a,$$

then $\rho = \eta$ λ_a-almost everywhere.

Proof. Equality of measures implies equality of their Radon–Nikodym derivatives with respect to λ_a. □

Definition 2.6 (Observational equivalence)**.** For $\rho, \eta \in \mathcal{D}_a$, define

$$\rho \sim_a \eta \iff P_{a,\rho}(q) = P_{a,\eta}(q) \qquad \forall q \in \mathcal{Q}_a.$$

Let

$$\widehat{\mathcal{D}}_a := \mathcal{D}_a / \sim_a.$$

Definition 2.7 (Admissible transport data)**.** Let $f : a \to b$ in $\mathcal{A}$. An *admissible transport datum* over f consists of:

1. a Markov kernel K_f from X_a to X_b;
2. a query pullback map
$$G_f : \mathcal{Q}_b \to \mathcal{Q}_a;$$
3. the absolute continuity condition
$$(K_f)_\#(\rho \, d\lambda_a) \ll \lambda_b \qquad \forall \rho \in \mathcal{D}_a;$$

4. the induced density transport

$$\mathsf{K}_f(\rho) := \frac{d\big((K_f)_\#(\rho\, d\lambda_a)\big)}{d\lambda_b} \in \mathcal{D}_b;$$

5. the compatibility relation

$$P_{b,\mathsf{K}_f(\rho)}(q) = P_{a,\rho}(G_f q) \qquad \forall \rho \in \mathcal{D}_a,\ \forall q \in \mathcal{Q}_b.$$

Lemma 2.8 (Well-defined transport on observational quotients)**.** *Let $f : a \to b$ carry admissible transport data. Then the assignment*

$$\Gamma_f : \widehat{\mathcal{D}}_a \to \widehat{\mathcal{D}}_b, \qquad \Gamma_f([\rho]_a) := [\mathsf{K}_f(\rho)]_b$$

is well-defined.

Proof. Assume $\rho \sim_a \eta$. Let $q \in \mathcal{Q}_b$. By compatibility,

$$P_{b,\mathsf{K}_f(\rho)}(q) = P_{a,\rho}(G_f q), \qquad P_{b,\mathsf{K}_f(\eta)}(q) = P_{a,\eta}(G_f q).$$

Since $G_f q \in \mathcal{Q}_a$ and $\rho \sim_a \eta$, one has

$$P_{a,\rho}(G_f q) = P_{a,\eta}(G_f q).$$

Hence

$$P_{b,\mathsf{K}_f(\rho)}(q) = P_{b,\mathsf{K}_f(\eta)}(q) \qquad \forall q \in \mathcal{Q}_b,$$

so $\mathsf{K}_f(\rho) \sim_b \mathsf{K}_f(\eta)$. □

Definition 2.9 (Observable-state functor)**.** Assume that for every $f : a \to b$ in $\mathcal{A}$, admissible transport data are chosen so that

$$\Gamma_{\mathrm{id}_a} = \mathrm{id}_{\widehat{\mathcal{D}}_a}, \qquad \Gamma_{g \circ f} = \Gamma_g \circ \Gamma_f.$$

Then the assignment

$$\widehat{\mathcal{D}} : \mathcal{A} \to \mathbf{Set}, \qquad a \mapsto \widehat{\mathcal{D}}_a, \quad f \mapsto \Gamma_f$$

is a covariant functor.

Definition 2.10 (Category of elements and discrete opfibration)**.** Define the category of elements

$$\int \widehat{\mathcal{D}}$$

as follows:

1. objects are pairs (a, ξ) with $a \in \mathrm{Ob}(\mathcal{A})$ and $\xi \in \widehat{\mathcal{D}}_a$;
2. a morphism

$$(a, \xi) \to (b, \zeta)$$

is a morphism $f : a \to b$ in $\mathcal{A}$ such that

$$\Gamma_f(\xi) = \zeta.$$

The projection

$$\pi : \int \widehat{\mathcal{D}} \to \mathcal{A}, \qquad \pi(a, \xi) = a, \quad \pi(f) = f,$$

is the canonical discrete opfibration associated with the covariant functor $\widehat{\mathcal{D}}$.

Definition 2.11 (Compatible global section)**.** A *compatible global section* is a functor

$$s : \mathcal{A} \to \int \widehat{\mathcal{D}}$$

such that

$$\pi \circ s = \mathrm{id}_{\mathcal{A}}.$$

Equivalently, s assigns to each $a \in \mathrm{Ob}(\mathcal{A})$ an element $s(a) \in \widehat{\mathcal{D}}_a$ with

$$\Gamma_f(s(a)) = s(b) \qquad \forall f : a \to b.$$

Theorem 2.12 (Fixed-point obstruction to compatible sections)**.** *Assume there exists a loop*

$$a_0 \xrightarrow{f_1} a_1 \xrightarrow{f_2} \cdots \xrightarrow{f_n} a_n = a_0$$

in $\mathcal{A}$, and let

$$\Gamma_\gamma := \Gamma_{f_n} \circ \cdots \circ \Gamma_{f_1} : \widehat{\mathcal{D}}_{a_0} \to \widehat{\mathcal{D}}_{a_0}.$$

If Γ_γ has no fixed point, then no compatible global section exists on the full subcategory generated by the loop.

Proof. If s were a compatible global section, then

$$\Gamma_\gamma(s(a_0)) = s(a_0)$$

by functoriality and $\pi \circ s = \mathrm{id}_{\mathcal{A}}$. This contradicts the absence of fixed points. □

Corollary 2.13 (Relative obstruction)**.** *Let $\xi \in \widehat{\mathcal{D}}_{a_0}$. If $\Gamma_\gamma(\xi) \neq \xi$, then no compatible global section s on the loop-generated subcategory can satisfy $s(a_0) = \xi$.*

Proof. If $s(a_0) = \xi$, compatibility along the loop forces $\Gamma_\gamma(\xi) = \xi$. □

3 Physical observer-states and exact semantic identity

Definition 3.1 (Physical observer-state)**.** A *physical observer-state* is a tuple

$$o = (a, X_o, \Sigma_o, \lambda_o, \rho_o, \mathcal{G}_o, \tau_o, \mathcal{Q}_o, r_o),$$

where:

1. $a \in \mathrm{Ob}(\mathcal{A})$ is a neighborhood of agency;
2. $X_o = X_S \times X_I \times X_E$ is the joint state space of system, instrument, and environment;
3. $(X_o, \Sigma_o, \lambda_o)$ is a measurable reference space;
4. ρ_o is a normalized joint density on X_o;

5. $\mathcal{G}_o \subseteq \Sigma_o$ is the accessible σ-algebra;

6. $\tau_o : X_o \to S$ is a semantic map;

7. $\mathcal{Q}_o \subseteq L^\infty(S)$ is a family of admissible semantic queries;

8. r_o is the effective resource level.

Define the associated density fibre

$$\mathcal{D}_o := \left\{ \rho \in L^1(X_o, \lambda_o) : \rho \geq 0, \int_{X_o} \rho \, d\lambda_o = 1 \right\},$$

so that $\rho_o \in \mathcal{D}_o$. Define

$$\nu_o := (\tau_o)_\#(\rho_o \, d\lambda_o) \in \mathcal{P}(S),$$

and, for $q \in \mathcal{Q}_o$,

$$P_o(q) := \int_S q \, d\nu_o = \int_{X_o} q(\tau_o(x)) \, \rho_o(x) \, d\lambda_o(x).$$

Definition 3.2 (Semantic profile). Let $\mathcal{Q}$ be a family of admissible semantic queries on a class $\mathcal{C} \subseteq \mathcal{O}$. Define

$$\Phi_\mathcal{Q}(o) := \big(P_o(q)\big)_{q \in \mathcal{Q}}.$$

Definition 3.3 (Separating family). A family $\mathcal{Q}$ is *separating* on $\mathcal{C} \subseteq \mathcal{O}$ if

$$\Phi_\mathcal{Q}(o) = \Phi_\mathcal{Q}(o') \Longrightarrow o \sim_g o'$$

for a chosen trivial equivalence relation $\sim_g$.

Theorem 3.4 (Exact semantic coincidence implies observer identity). *Let $\mathcal{Q}$ be separating on $\mathcal{C} \subseteq \mathcal{O}$. Then*

$$\Phi_\mathcal{Q}(o) = \Phi_\mathcal{Q}(o') \Longrightarrow o \sim_g o'.$$

Proof. Immediate from the definition of a separating family. □

4 Agreement radius and non-compressibility

Definition 4.1 (Observer pseudometric and semantic discrepancy). Let d_{obs} be a pseudometric on $\mathcal{O}$. Let d_{sem} be a metric or pseudometric on semantic profile space. For $\varepsilon > 0$, define

$$A_\varepsilon(o) := \{o' \in \mathcal{O} : \ d_{\text{sem}}(\Phi_\mathcal{Q}(o), \Phi_\mathcal{Q}(o')) \leq \varepsilon\}.$$

Define the finite-distance component of o by

$$\mathcal{C}(o) := \{o' \in \mathcal{O} : \ d_{\text{obs}}(o, o') < \infty\}.$$

Definition 4.2 (Flattenability threshold). For $\mathcal{C} \subseteq \mathcal{O}$, define

$$\varepsilon_{\text{flat}}(\mathcal{C}) := \inf \left\{ \varepsilon > 0 : \ \exists o_* \in \mathcal{C} \text{ such that } \sup_{o \in \mathcal{C}} d_{\text{sem}}(\Phi_\mathcal{Q}(o), \Phi_\mathcal{Q}(o_*)) \leq \varepsilon \right\}.$$

Definition 4.3 (Agreement radius). For $o \in \mathcal{O}$, define

$$R_\varepsilon(o) := \sup\{r > 0 : \ B_r^{\text{obs}}(o) \subseteq A_\varepsilon(o)\}.$$

Hypothesis 4.4 (Local semantic continuity)**.** The profile map $\Phi_{\mathcal{Q}}$ is locally continuous from $(\mathcal{O}, d_{\text{obs}})$ to semantic profile space equipped with d_{sem}.

Theorem 4.5 (Finite nonzero agreement radius)**.** *Assume local semantic continuity. Let $o \in \mathcal{O}$, and suppose*

$$\varepsilon < \varepsilon_{\text{flat}}(\mathcal{C}(o)).$$

Then

$$0 < R_\varepsilon(o) < \infty.$$

Proof. By local continuity there exists $\delta > 0$ such that

$$d_{\text{obs}}(o, o') < \delta \Longrightarrow d_{\text{sem}}(\Phi_{\mathcal{Q}}(o), \Phi_{\mathcal{Q}}(o')) \leq \varepsilon.$$

Hence $B_\delta^{\text{obs}}(o) \subseteq A_\varepsilon(o)$, so $R_\varepsilon(o) \geq \delta > 0$.

If $R_\varepsilon(o) = \infty$, then every $o' \in \mathcal{C}(o)$ lies in some finite ball $B_r^{\text{obs}}(o)$, hence in $A_\varepsilon(o)$. Therefore

$$\sup_{o' \in \mathcal{C}(o)} d_{\text{sem}}(\Phi_{\mathcal{Q}}(o), \Phi_{\mathcal{Q}}(o')) \leq \varepsilon,$$

contradicting $\varepsilon < \varepsilon_{\text{flat}}(\mathcal{C}(o))$. □

Definition 4.6 (Collapsing compression family)**.** Let $\mathcal{C} \subseteq \mathcal{O}$. A family

$$C_\lambda : \mathcal{C} \to \mathcal{O}, \qquad \lambda \downarrow 0,$$

is *collapsing on $\mathcal{C}$* if

$$\operatorname{diam}_{d_{\text{obs}}}(C_\lambda(\mathcal{C})) \to 0.$$

Definition 4.7 (Semantically lossless compression)**.** A collapsing family C_λ is *semantically lossless on $\mathcal{C}$* if

$$\sup_{o \in \mathcal{C}} d_{\text{sem}}(\Phi_{\mathcal{Q}}(C_\lambda o), \Phi_{\mathcal{Q}}(o)) \to 0 \qquad (\lambda \downarrow 0).$$

Hypothesis 4.8 (Uniform continuity on compressed images)**.** For every $\varepsilon > 0$, there exists $\delta > 0$ such that for all sufficiently small λ, and all $x, y \in C_\lambda(\mathcal{C})$,

$$d_{\text{obs}}(x, y) < \delta \Longrightarrow d_{\text{sem}}(\Phi_{\mathcal{Q}}(x), \Phi_{\mathcal{Q}}(y)) < \varepsilon.$$

Theorem 4.9 (No semantically lossless collapsing compression above threshold)**.** *Let $\mathcal{C} \subseteq \mathcal{O}$ satisfy $\varepsilon_{\text{flat}}(\mathcal{C}) > 0$. Under the uniform continuity hypothesis, there exists no family C_λ that is both collapsing and semantically lossless on $\mathcal{C}$.*

Proof. Choose $0 < \varepsilon < \varepsilon_{\text{flat}}(\mathcal{C})/3$. By semantic losslessness, for sufficiently small λ,

$$\sup_{o \in \mathcal{C}} d_{\text{sem}}(\Phi_{\mathcal{Q}}(C_\lambda o), \Phi_{\mathcal{Q}}(o)) < \varepsilon.$$

By collapse and uniform continuity, for sufficiently small λ,

$$\sup_{x, y \in C_\lambda(\mathcal{C})} d_{\text{sem}}(\Phi_{\mathcal{Q}}(x), \Phi_{\mathcal{Q}}(y)) < \varepsilon.$$

Fix such a λ, and choose $o_* \in \mathcal{C}$. Then for all $o \in \mathcal{C}$,

$$\begin{aligned} d_{\text{sem}}(\Phi_{\mathcal{Q}}(o), \Phi_{\mathcal{Q}}(o_*)) &\leq d_{\text{sem}}(\Phi_{\mathcal{Q}}(o), \Phi_{\mathcal{Q}}(C_\lambda o)) + d_{\text{sem}}(\Phi_{\mathcal{Q}}(C_\lambda o), \Phi_{\mathcal{Q}}(C_\lambda o_*)) + d_{\text{sem}}(\Phi_{\mathcal{Q}}(C_\lambda o_*), \Phi_{\mathcal{Q}}(o_*)) \\ &< 3\varepsilon < \varepsilon_{\text{flat}}(\mathcal{C}), \end{aligned}$$

contradiction. □

5 Protocol category, semantic realization, abstract holonomy, and realized path residue

5.1 Protocol category

Definition 5.1 (Universe of queries)**.** Fix a global syntactic query universe $\mathcal{Q}^*$. For every protocol state π, its admissible query family satisfies

$$\mathcal{Q}_\pi \subseteq \mathcal{Q}^*.$$

Definition 5.2 (Resource category)**.** A *resource category* **Res** is a thin category whose objects are resource levels. A protocol resource r_π is an object of **Res**, and a resource update $r_\gamma : r_\pi \to r_{\pi'}$ is a morphism in **Res**.

Definition 5.3 (Protocol state)**.** A protocol state is a tuple

$$\pi = (o, X_\pi, \Sigma_\pi, \lambda_\pi, \rho_\pi, \mathcal{Q}_\pi, \mathcal{P}_\pi, \mathcal{R}_\pi, \mathcal{C}_\pi, r_\pi),$$

where:

1. $o \in \mathcal{O}$ is an observer-state;
2. $X_\pi := X_o$, $\Sigma_\pi := \Sigma_o$, $\lambda_\pi := \lambda_o$, and $\rho_\pi := \rho_o$;
3. $\mathcal{Q}_\pi := \mathcal{Q}_o \subseteq \mathcal{Q}^*$;
4. $\mathcal{P}_\pi$ is a preparation rule, $\mathcal{R}_\pi$ a readout rule, and $\mathcal{C}_\pi$ a calibration, compression, or extrapolation rule;
5. $r_\pi \in \mathrm{Ob}(\mathbf{Res})$ is a resource object. The default convention is $r_\pi := r_o$.

Definition 5.4 (Protocol category)**.** The category **Prot** has protocol states as objects. A morphism

$$\gamma : \pi \to \pi'$$

is a triple

$$\gamma = (K_\gamma, G_\gamma, r_\gamma),$$

where:

1. $K_\gamma : \mathcal{P}(X_\pi) \to \mathcal{P}(X_{\pi'})$ is a stochastic state transport;
2. $G_\gamma : \mathcal{Q}_{\pi'} \to \mathcal{Q}_\pi$ is a query pullback map;
3. $r_\gamma : r_\pi \to r_{\pi'}$ is a morphism in **Res**;
4. the pullback is norm-bounded:

$$L_\gamma := \sup_{q \in \mathcal{Q}_{\pi'}} \frac{1 + \|G_\gamma q\|_\infty}{1 + \|q\|_\infty} < \infty;$$

5. the semantic compatibility condition holds:

$$P_{\pi'}(q) = P_\pi(G_\gamma q) \qquad \forall q \in \mathcal{Q}_{\pi'}.$$

For composable $\gamma : \pi \to \pi'$ and $\gamma' : \pi' \to \pi''$, define

$$\gamma' \circ \gamma := (K_{\gamma'} \circ K_\gamma,\ G_\gamma \circ G_{\gamma'},\ r_{\gamma'} \circ r_\gamma),$$

where the second component is the usual contravariant composition of pullback maps. The identity at π is

$$\mathrm{id}_\pi := (\mathrm{id}, \mathrm{id}, \mathrm{id}).$$

The invertible morphisms form a subgroupoid

$$\mathbf{InvProt} \subseteq \mathbf{Prot}.$$

5.2 Semantic realization and abstract holonomy

Definition 5.5 (Semantic realization category)**.** Let **Sem** be the category whose objects are pointed pseudometric spaces

$$(\mathfrak{S}, d, \mathbf{s}),$$

and whose morphisms are basepoint-preserving Lipschitz maps.

For each protocol state π, define

$$\mathfrak{S}_\pi := \{u : \mathcal{Q}_\pi \to \mathbb{R} : \|u\|_\pi < \infty\}, \qquad \|u\|_\pi := \sup_{q \in \mathcal{Q}_\pi} \frac{|u(q)|}{1 + \|q\|_\infty},$$

with pseudometric

$$d_\pi(u, v) := \|u - v\|_\pi.$$

Define the distinguished semantic point

$$\Phi_\pi \in \mathfrak{S}_\pi, \qquad \Phi_\pi(q) := P_\pi(q).$$

Definition 5.6 (Semantic transport functor)**.** A semantic transport functor is a functor

$$\mathfrak{F} : \mathbf{Prot} \to \mathbf{Sem}$$

such that, for every protocol state π,

$$\mathfrak{F}(\pi) = (\mathfrak{S}_\pi, d_\pi, \Phi_\pi),$$

and for every morphism $\gamma : \pi \to \pi'$,

$$T_\gamma := \mathfrak{F}(\gamma) : \mathfrak{S}_\pi \to \mathfrak{S}_{\pi'}$$

is given by

$$(T_\gamma u)(q) := u(G_\gamma q), \qquad q \in \mathcal{Q}_{\pi'}.$$

Lemma 5.7 (Well-defined semantic transport)**.** *For every protocol morphism $\gamma : \pi \to \pi'$, the map*

$$T_\gamma : \mathfrak{S}_\pi \to \mathfrak{S}_{\pi'}$$

is well-defined, Lipschitz with constant at most L_γ, and basepoint-preserving:

$$T_\gamma(\Phi_\pi) = \Phi_{\pi'}.$$

Proof. Let $u \in \mathfrak{S}_\pi$. Then for $q \in \mathcal{Q}_{\pi'}$,

$$\frac{|(T_\gamma u)(q)|}{1+\|q\|_\infty} = \frac{|u(G_\gamma q)|}{1+\|q\|_\infty} \le \|u\|_\pi \, \frac{1+\|G_\gamma q\|_\infty}{1+\|q\|_\infty}.$$

Taking the supremum over $q \in \mathcal{Q}_{\pi'}$ yields

$$\|T_\gamma u\|_{\pi'} \le L_\gamma \|u\|_\pi < \infty,$$

so $T_\gamma u \in \mathfrak{S}_{\pi'}$. Similarly,

$$d_{\pi'}(T_\gamma u, T_\gamma v) \le L_\gamma d_\pi(u,v).$$

Finally, for $q \in \mathcal{Q}_{\pi'}$,

$$(T_\gamma \Phi_\pi)(q) = \Phi_\pi(G_\gamma q) = P_\pi(G_\gamma q) = P_{\pi'}(q) = \Phi_{\pi'}(q),$$

by semantic compatibility. □

Lemma 5.8 (Functoriality of semantic transport)**.** *For every protocol state π and every composable pair of protocol morphisms*

$$\pi \xrightarrow{\gamma} \pi' \xrightarrow{\gamma'} \pi'',$$

one has

$$T_{\mathrm{id}_\pi} = \mathrm{id}_{\mathfrak{S}_\pi} \qquad \textit{and} \qquad T_{\gamma' \circ \gamma} = T_{\gamma'} \circ T_\gamma.$$

Hence the assignment $\mathfrak{F}$ of Definition 5.6 is indeed a functor $\mathbf{Prot} \to \mathbf{Sem}$*.*

Proof. For $u \in \mathfrak{S}_\pi$ and $q \in \mathcal{Q}_\pi$,

$$(T_{\mathrm{id}_\pi} u)(q) = u(G_{\mathrm{id}_\pi} q) = u(q),$$

so $T_{\mathrm{id}_\pi} = \mathrm{id}_{\mathfrak{S}_\pi}$. For composable γ, γ' and $q \in \mathcal{Q}_{\pi''}$,

$$(T_{\gamma'\circ\gamma} u)(q) = u\big(G_{\gamma'\circ\gamma} q\big) = u\big((G_\gamma \circ G_{\gamma'}) q\big) = (T_\gamma u)(G_{\gamma'} q) = (T_{\gamma'}(T_\gamma u))(q),$$

where the middle identity uses the definition of composition in **Prot**. Thus $T_{\gamma'\circ\gamma} = T_{\gamma'} \circ T_\gamma$. □

Definition 5.9 (Loop and abstract observational holonomy)**.** A loop at π is a composable sequence

$$\ell = \gamma_n \circ \cdots \circ \gamma_1$$

whose source and target are both π. Its abstract observational holonomy is

$$\mathrm{Hol}_{\mathrm{abs}}(\ell;\pi) := d_\pi(T_\ell(\Phi_\pi), \Phi_\pi)\,.$$

Proposition 5.10 (Path-independence implies vanishing abstract holonomy)**.** *If $T_p = T_q$ for any two composable paths $p, q : \pi \to \pi'$, then every loop at every protocol state has zero abstract observational holonomy.*

Proof. Let ℓ be a loop at π. By path-independence, the transport along ℓ agrees with the transport along the identity path at π, so

$$T_\ell = T_{\mathrm{id}_\pi}.$$

Hence

$$\mathrm{Hol}_{\mathrm{abs}}(\ell;\pi) = d_\pi\big(T_\ell(\Phi_\pi), \Phi_\pi\big) = d_\pi\big(T_{\mathrm{id}_\pi}(\Phi_\pi), \Phi_\pi\big) = d_\pi(\Phi_\pi, \Phi_\pi) = 0.$$

□

5.3 Realized paths and path modulus

Definition 5.11 (Realized path)**.** A realized path from π_0 to π_n is a finite composable chain

$$p:\ \pi_0 \xrightarrow{\gamma_1} \pi_1 \xrightarrow{\gamma_2} \cdots \xrightarrow{\gamma_n} \pi_n$$

in **Prot**. The path p is *contained in* $B_r^{\mathrm{prot}}(\pi_0)$ if

$$\pi_i \in B_r^{\mathrm{prot}}(\pi_0) \qquad \forall i = 0, \dots, n.$$

Definition 5.12 (Realized path residue)**.** Let

$$p:\ \pi_0 \xrightarrow{\gamma_1} \pi_1 \xrightarrow{\gamma_2} \cdots \xrightarrow{\gamma_n} \pi_n$$

be a realized path. Define its common comparison frame by

$$\mathcal{Q}(p) := \bigcap_{i=0}^{n} \mathcal{Q}_{\pi_i} \subseteq \mathcal{Q}^*.$$

Its realized residue relative to the initial state is

$$\mathrm{Res}_{\mathrm{obs}}(p;\pi_0) := \sup_{q\in\mathcal{Q}(p)} \frac{|P_{\pi_n}(q) - P_{\pi_0}(q)|}{1+\|q\|_\infty}.$$

If $\mathcal{Q}(p) = \varnothing$, this supremum is 0 by convention.

Definition 5.13 (Path modulus)**.** For a protocol state π, define

$$\mathfrak{P}_\pi(r) := \sup\left\{\mathrm{Res}_{\mathrm{obs}}(p;\pi) : p \text{ is a realized path starting at } \pi,\ p \subseteq B_r^{\mathrm{prot}}(\pi)\right\}.$$

Definition 5.14 (Ball-framed realized path residue)**.** Let π be a protocol state, let $r \geq 0$, and let

$$p:\ \pi = \pi_0 \xrightarrow{\gamma_1} \pi_1 \xrightarrow{\gamma_2} \cdots \xrightarrow{\gamma_n} \pi_n$$

be a realized path contained in $B_r^{\mathrm{prot}}(\pi)$. Its *ball-framed realized residue* is

$$\widetilde{\mathrm{Res}}_{\mathrm{obs}}(p;\pi,r) := \sup_{q\in\mathcal{Q}_{\pi,r}} \frac{|P_{\pi_n}(q) - P_{\pi}(q)|}{1+\|q\|_\infty}.$$

If $\mathcal{Q}_{\pi,r} = \varnothing$, this supremum is 0 by convention.

Definition 5.15 (Ball-framed path modulus)**.** For a protocol state π, define

$$\widetilde{\mathfrak{P}}_\pi(r) := \sup\left\{\widetilde{\mathrm{Res}}_{\mathrm{obs}}(p;\pi,r) : p \text{ is a realized path starting at } \pi,\ p \subseteq B_r^{\mathrm{prot}}(\pi)\right\}.$$

Remark 5.16 (Two path residues)*.* The realized residue $\mathrm{Res}_{\mathrm{obs}}(p;\pi_0)$ measures the semantic discrepancy intrinsic to the common path frame $\mathcal{Q}(p)$. The ball-framed residue $\widetilde{\mathrm{Res}}_{\mathrm{obs}}(p;\pi,r)$ is the version compatible with the agreement modulus $\mathfrak{E}_\pi(r)$, because both are measured in the same comparison frame $\mathcal{Q}_{\pi,r}$.

Theorem 5.17 (Ball-framed path modulus upper bound on agreement radius)**.** *If*

$$\widetilde{\mathfrak{P}}_\pi(r) > \varepsilon,$$

then

$$R_\varepsilon(\pi) < r.$$

Proof. If $R_\varepsilon(\pi) \geq r$, then by definition of $\mathfrak{E}_\pi(r)$, every protocol state $\pi' \in B_r^{\mathrm{prot}}(\pi)$ satisfies

$$\sup_{q \in \mathcal{Q}_{\pi,r}} \frac{|P_{\pi'}(q) - P_\pi(q)|}{1 + \|q\|_\infty} \leq \varepsilon.$$

Hence, for every realized path p starting at π and contained in $B_r^{\mathrm{prot}}(\pi)$, its endpoint π_n also lies in $B_r^{\mathrm{prot}}(\pi)$, so

$$\widetilde{\mathrm{Res}}_{\mathrm{obs}}(p; \pi, r) \leq \varepsilon.$$

Taking the supremum over such paths gives $\widetilde{\mathfrak{P}}_\pi(r) \leq \varepsilon$, contradiction. $\square$

Definition 5.18 (Reconstructible compression loops)**.** A family of compression morphisms $\chi_\lambda : \pi \to \pi_\lambda$ is *semantically reconstructible* if there exist reconstruction morphisms $\eta_\lambda : \pi_\lambda \to \pi$ and a modulus $\psi : [0, \infty) \to [0, \infty)$ with $\psi(t) \to 0$ as $t \downarrow 0$, such that

$$d_\pi\Big((T_{\eta_\lambda} \circ T_{\chi_\lambda})(\Phi_\pi), \Phi_\pi\Big) \leq \psi\left(\sup_{q \in \mathcal{Q}_\pi \cap \mathcal{Q}_{\pi_\lambda}} \frac{|P_{\pi_\lambda}(q) - P_\pi(q)|}{1 + \|q\|_\infty}\right).$$

Equivalently, by Lemma 5.8,

$$d_\pi\Big(T_{\eta_\lambda \circ \chi_\lambda}(\Phi_\pi), \Phi_\pi\Big) \leq \psi\left(\sup_{q \in \mathcal{Q}_\pi \cap \mathcal{Q}_{\pi_\lambda}} \frac{|P_{\pi_\lambda}(q) - P_\pi(q)|}{1 + \|q\|_\infty}\right).$$

Theorem 5.19 (Positive compression residue forbids pointwise semantically lossless reconstructible compression)**.** *Suppose χ_λ is semantically reconstructible and pointwise semantically lossless at π, i.e.*

$$\sup_{q \in \mathcal{Q}_\pi \cap \mathcal{Q}_{\pi_\lambda}} \frac{|P_{\pi_\lambda}(q) - P_\pi(q)|}{1 + \|q\|_\infty} \to 0.$$

Then

$$\mathrm{Hol}_{\mathrm{abs}}(\eta_\lambda \circ \chi_\lambda; \pi) \to 0.$$

Consequently, if there exists $c > 0$ and a sequence $\lambda_n \downarrow 0$ such that

$$\mathrm{Hol}_{\mathrm{abs}}(\eta_{\lambda_n} \circ \chi_{\lambda_n}; \pi) \geq c \qquad \forall n,$$

then no such pointwise semantically lossless reconstructible compression exists.

Proof. The first statement follows immediately from the reconstructibility modulus ψ. The second is its contrapositive. $\square$

6 Agreement modulus and finite-bank invariants

Definition 6.1 (Agreement modulus)**.** For a protocol state π and $r \geq 0$, define the common comparison frame

$$\mathcal{Q}_{\pi,r} := \bigcap_{\pi' : \, d_{\mathrm{prot}}(\pi, \pi') \leq r} \mathcal{Q}_{\pi'} \subseteq \mathcal{Q}^*.$$

Then define

$$\mathfrak{E}_\pi(r) := \sup\left\{\sup_{q \in \mathcal{Q}_{\pi,r}} \frac{|P_{\pi'}(q) - P_\pi(q)|}{1 + \|q\|_\infty} : \ d_{\mathrm{prot}}(\pi, \pi') \leq r\right\}.$$

If $\mathcal{Q}_{\pi,r} = \varnothing$, the inner supremum is 0 by convention.

Definition 6.2 (Local overlap continuity)**.** We say that protocol semantics is *locally overlap-continuous at* π if there exist $r_{\mathrm{loc}} > 0$ and a nondecreasing modulus $\omega_\pi : [0,\infty) \to [0,\infty)$, with $\omega_\pi(t) \to 0$ as $t \downarrow 0$, such that for every π' with

$$d_{\mathrm{prot}}(\pi, \pi') \le r_{\mathrm{loc}},$$

one has

$$\sup_{q \in \mathcal{Q}_\pi \cap \mathcal{Q}_{\pi'}} \frac{|P_{\pi'}(q) - P_\pi(q)|}{1 + \|q\|_\infty} \le \omega_\pi(d_{\mathrm{prot}}(\pi, \pi')).$$

Proposition 6.3 (Basic properties of the agreement modulus)**.** *For every* π*,*

$$\mathfrak{E}_\pi(0) = 0, \qquad r_1 \le r_2 \Longrightarrow \mathfrak{E}_\pi(r_1) \le \mathfrak{E}_\pi(r_2).$$

If protocol semantics is locally overlap-continuous at π*, then*

$$\mathfrak{E}_\pi(r) \to 0 \qquad (r \downarrow 0).$$

More precisely, for $0 \le r \le r_{\mathrm{loc}}$*,*

$$\mathfrak{E}_\pi(r) \le \omega_\pi(r).$$

Proof. The first two claims are immediate from the definition and monotonicity of balls. For the local estimate, let $0 \le r \le r_{\mathrm{loc}}$ and let π' satisfy $d_{\mathrm{prot}}(\pi, \pi') \le r$. Since

$$\mathcal{Q}_{\pi,r} \subseteq \mathcal{Q}_\pi \cap \mathcal{Q}_{\pi'},$$

one has

$$\sup_{q \in \mathcal{Q}_{\pi,r}} \frac{|P_{\pi'}(q) - P_\pi(q)|}{1 + \|q\|_\infty} \le \sup_{q \in \mathcal{Q}_\pi \cap \mathcal{Q}_{\pi'}} \frac{|P_{\pi'}(q) - P_\pi(q)|}{1 + \|q\|_\infty} \le \omega_\pi(d_{\mathrm{prot}}(\pi, \pi')) \le \omega_\pi(r).$$

Taking the outer supremum over all such π' yields

$$\mathfrak{E}_\pi(r) \le \omega_\pi(r).$$

Since $\omega_\pi(r) \to 0$ as $r \downarrow 0$, the conclusion follows. □

Definition 6.4 (Agreement radius as inverse modulus)**.**

$$R_\varepsilon(\pi) := \sup\{r > 0 : \ \mathfrak{E}_\pi(r) \le \varepsilon\}.$$

Definition 6.5 (Upper semantic dilatation)**.**

$$\mathrm{Dil}^+_{\mathcal{Q}}(\pi) := \limsup_{r \downarrow 0} \frac{\mathfrak{E}_\pi(r)}{r}.$$

Proposition 6.6 (Local lower bound)**.** *If there exist* $L_\pi < \infty$ *and* $r_0 > 0$ *such that*

$$\mathfrak{E}_\pi(r) \le L_\pi r \qquad (0 \le r \le r_0),$$

then

$$R_\varepsilon(\pi) \ge \min\left(r_0, \frac{\varepsilon}{L_\pi}\right).$$

Proof. If $r \le r_0$ and $r \le \varepsilon / L_\pi$, then $\mathfrak{E}_\pi(r) \le \varepsilon$. □

Definition 6.7 (Finite protocol bank)**.** A finite protocol bank rooted at π is a finite set $B_N(\pi) \subseteq \mathrm{Ob}(\mathbf{Prot})$ such that $\pi \in B_N(\pi)$.

Definition 6.8 (Bank-admissible finite query family)**.** Let $B_N(\pi)$ be a rooted finite protocol bank. A finite family

$$\mathcal{Q}_N(\pi)$$

is *bank-admissible* if

$$\mathcal{Q}_N(\pi) \subseteq \bigcap_{\pi' \in B_N(\pi)} \mathcal{Q}_{\pi'} \subseteq \mathcal{Q}^*.$$

It is *bank-separating* if the map

$$\pi' \longmapsto \big(P_{\pi'}(q)\big)_{q \in \mathcal{Q}_N(\pi)}$$

is injective on $B_N(\pi)$ modulo the chosen trivial equivalence relation.

Definition 6.9 (Finite-bank estimator)**.** Let $B_N(\pi)$ be a rooted finite protocol bank and let $\mathcal{Q}_N(\pi)$ be a bank-admissible finite query family. Define

$$\widehat{d}_{\mathrm{sem},N}(\pi,\pi') := \sup_{q \in \mathcal{Q}_N(\pi)} |P_\pi(q) - P_{\pi'}(q)|,$$

$$\widehat{\mathfrak{E}}_{\pi,N}(r) := \max\left\{\widehat{d}_{\mathrm{sem},N}(\pi,\pi') : \pi' \in B_N(\pi),\ d_{\mathrm{prot}}(\pi,\pi') \le r\right\},$$

$$\widehat{R}_{\varepsilon,N}(\pi) := \sup\{r > 0 :\ \widehat{\mathfrak{E}}_{\pi,N}(r) \le \varepsilon\}.$$

Remark 6.10 (Normalization convention for the finite estimator)**.** The continuous modulus $\mathfrak{E}_\pi(r)$ is normalized by the factor $(1+\|q\|_\infty)^{-1}$, whereas the finite-bank estimator $\widehat{d}_{\mathrm{sem},N}$ is left unnormalized. This asymmetry is intentional: Hypothesis 6.10 is understood to absorb the resulting comparison constants between the finite bank and the continuous modulus on each fixed window.

Hypothesis 6.11 (Transfer modulus)**.** There exists $\omega_N \to 0$ such that for every finite window $0 \le r \le r_*$,

$$\sup_{0 \le r \le r_*} \left|\widehat{\mathfrak{E}}_{\pi,N}(r) - \mathfrak{E}_\pi(r)\right| \le \omega_N.$$

Definition 6.12 (Isolated threshold crossing)**.** Fix $\varepsilon > 0$. We say that $\mathfrak{E}_\pi$ has an *isolated threshold crossing* at $R_\varepsilon(\pi)$ if there exists $\delta_0 > 0$ such that for every $0 < \delta < \delta_0$, whenever $R_\varepsilon(\pi) \pm \delta$ lie in the window under consideration,

$$\mathfrak{E}_\pi(R_\varepsilon(\pi) - \delta) < \varepsilon \quad \text{and} \quad \mathfrak{E}_\pi(R_\varepsilon(\pi) + \delta) > \varepsilon.$$

Theorem 6.13 (Consistency of empirical agreement radius)**.** *Assume the transfer modulus hypothesis and an isolated threshold crossing at $R_\varepsilon(\pi)$. Then*

$$\widehat{R}_{\varepsilon,N}(\pi) \to R_\varepsilon(\pi) \qquad (N \to \infty).$$

Proof. Fix $\delta \in (0, \delta_0)$. By isolated crossing,

$$\mathfrak{E}_\pi(R_\varepsilon(\pi) - \delta) < \varepsilon < \mathfrak{E}_\pi(R_\varepsilon(\pi) + \delta).$$

Let

$$\alpha_\delta := \varepsilon - \mathfrak{E}_\pi(R_\varepsilon(\pi) - \delta) > 0, \qquad \beta_\delta := \mathfrak{E}_\pi(R_\varepsilon(\pi) + \delta) - \varepsilon > 0.$$

Choose N so that $\omega_N < \min(\alpha_\delta, \beta_\delta)$. Then

$$\widehat{\mathfrak{E}}_{\pi,N}(R_\varepsilon(\pi) - \delta) < \varepsilon, \qquad \widehat{\mathfrak{E}}_{\pi,N}(R_\varepsilon(\pi) + \delta) > \varepsilon.$$

Hence

$$R_\varepsilon(\pi) - \delta \le \widehat{R}_{\varepsilon,N}(\pi) \le R_\varepsilon(\pi) + \delta.$$

Since $\delta > 0$ is arbitrary, the conclusion follows. $\square$

7 Order induced by non-closure

Definition 7.1 (Obstruction vector)**.** Let Z be a class of semantic states. For each resource level r, fix five nonnegative obstruction functionals

$$\kappa_r, \qquad \mathrm{Hol}_r^{\mathrm{atlas}}, \qquad \mathrm{Hol}_r^{\mathrm{prot}}, \qquad \mathfrak{g}_r, \qquad \mathfrak{c}_r$$

on Z. Define

$$\Delta_r(z) = \big(\kappa_r(z),\ \mathrm{Hol}_r^{\mathrm{atlas}}(z),\ \mathrm{Hol}_r^{\mathrm{prot}}(z),\ \mathfrak{g}_r(z),\ \mathfrak{c}_r(z)\big) \in [0,\infty)^5.$$

Definition 7.2 (Refinement preorder)**.** For $x, y \in Z$, define

$$x \preceq_r y \iff \Delta_r(y) \le \Delta_r(x) \quad \text{componentwise.}$$

Define

$$x \sim_r y \iff x \preceq_r y \text{ and } y \preceq_r x.$$

Proposition 7.3. *The relation $\preceq_r$ is a preorder, and*

$$x \sim_r y \iff \Delta_r(x) = \Delta_r(y).$$

Proof. Immediate from the componentwise order on $[0,\infty)^5$. □

Definition 7.4 (Global trivialization)**.** A class Z is globally trivialized at scale r if there exists a global chart, section, or normalization on Z such that

$$\Delta_r(z) = 0 \qquad \forall z \in Z.$$

Proposition 7.5 (Global trivialization collapses the quotient)**.** *If Z is globally trivialized at scale r, then $Z/{\sim_r}$ has a single class.*

Proof. If $\Delta_r(z) = 0$ for all z, then $\Delta_r(x) = \Delta_r(y)$ for all x, y, hence $x \sim_r y$. □

Theorem 7.6 (Persistent obstruction prevents global trivialization)**.** *If*

$$\inf_{z \in Z} \|\Delta_r(z)\|_\infty > 0,$$

then Z is not globally trivialized at scale r.

Proof. Global trivialization would imply $\Delta_r(z) = 0$ for all z. □

Theorem 7.7 (Non-constant obstruction prevents collapse)**.** *If Δ_r is not constant on Z, then $Z/{\sim_r}$ has at least two distinct classes.*

Proof. If Δ_r is not constant, choose $x, y \in Z$ with $\Delta_r(x) \neq \Delta_r(y)$. Then $x \not\sim_r y$. □

Definition 7.8 (Deployment path)**.** A deployment path at scale r is a chain

$$z_0 \preceq_r z_1 \preceq_r \cdots \preceq_r z_n.$$

8 Explicit binary model

8.1 Observer-protocol state

Let $S \in \{0, 1\}$ be a latent variable with prevalence

$$p := \mathbb{P}(S = 1) \in [0, 1].$$

An instrument produces $M \in \{0, 1\}$ with false positive and false negative rates

$$\mathbb{P}(M = 1 \mid S = 0) = \alpha, \qquad \mathbb{P}(M = 0 \mid S = 1) = \beta.$$

Define the observer-protocol state

$$\pi = (\alpha, \beta, p).$$

Fix $\eta > 0$ and define

$$D_\eta := \{(\alpha, \beta, p) \in [0, 1]^3 : \ \alpha + \beta \leq 1 - \eta\}.$$

Define the semantic queries

$$P_\pi(q_0) = \alpha, \qquad P_\pi(q_1) = 1 - \beta, \qquad P_\pi(q_*) = \alpha(1 - p) + (1 - \beta)p.$$

Hence

$$\Phi(\pi) = \big(\alpha,\ 1 - \beta,\ \theta(\alpha, \beta, p)\big),$$

where

$$\theta(\alpha, \beta, p) := \alpha(1 - p) + (1 - \beta)p = \alpha + p(1 - \alpha - \beta).$$

Theorem 8.1 (Exact identity in the binary model)**.** *If $\pi, \pi' \in D_\eta$ and*

$$\Phi(\pi) = \Phi(\pi'),$$

then $\pi = \pi'$.

Proof. Equality of the first two coordinates yields $\alpha = \alpha'$ and $\beta = \beta'$. Equality of the third gives

$$(1 - \alpha - \beta)p = (1 - \alpha - \beta)p'.$$

Since $1 - \alpha - \beta \geq \eta > 0$, one obtains $p = p'$. □

8.2 Agreement window

Define

$$d_{\text{prot}}(\pi, \pi') := \max\{|\alpha - \alpha'|, |\beta - \beta'|, |p - p'|\},$$

and

$$d_{\text{sem}}(\Phi(\pi), \Phi(\pi')) := \max\big\{|\alpha - \alpha'|, |\beta - \beta'|, |\theta(\alpha, \beta, p) - \theta(\alpha', \beta', p')|\big\}.$$

Proposition 8.2 (Local Lipschitz estimate)**.** *For all $\pi, \pi' \in D_\eta$,*

$$d_{\text{sem}}(\Phi(\pi), \Phi(\pi')) \leq 2\, d_{\text{prot}}(\pi, \pi').$$

Proof. The first two coordinates contribute at most $d_{\text{prot}}(\pi, \pi')$. For the third, the gradient of θ on the convex domain D_η satisfies

$$\left|\frac{\partial\theta}{\partial\alpha}\right| = 1 - p, \qquad \left|\frac{\partial\theta}{\partial\beta}\right| = p, \qquad \left|\frac{\partial\theta}{\partial p}\right| = 1 - \alpha - \beta.$$

Hence

$$\left|\frac{\partial\theta}{\partial\alpha}\right| + \left|\frac{\partial\theta}{\partial\beta}\right| + \left|\frac{\partial\theta}{\partial p}\right| \leq 2.$$

By the mean-value theorem on D_η,

$$|\theta(\pi) - \theta(\pi')| \leq 2\, d_{\text{prot}}(\pi, \pi').$$

Taking the maximum yields the claim. □

Definition 8.3 (Admissible local radius)**.** Define

$$r_{\text{adm}}(\pi) := \sup\{r > 0 : \; B_r^{\text{prot}}(\pi) \subseteq D_\eta\}.$$

Corollary 8.4 (Lower agreement bound)**.** *For every $\pi \in D_\eta$,*

$$R_\varepsilon(\pi) \geq \min\left(r_{\text{adm}}(\pi), \frac{\varepsilon}{2}\right).$$

Proof. Apply the local lower-bound proposition to the Lipschitz constant $L_\pi = 2$. □

8.3 Control words and realized path residue

Define concrete transformations

$$\gamma_u(\alpha, \beta, p) := (\alpha + u, \beta, p), \qquad \delta_v(\alpha, \beta, p) := (\alpha, \beta, p + v\alpha),$$

whenever the resulting states remain in D_η.

At the level of the explicit model, each concrete transformation is interpreted as a protocol morphism, and its realized semantic transport is defined by

$$T_\gamma^{\text{real}}(\Phi(\pi)) := \Phi(\gamma\pi)$$

for every concrete protocol transformation γ generated by γ_u, δ_v, their inverses when defined, and their compositions. For a concrete control word w, define its realized residue by

$$\operatorname{Res}_{\text{obs}}(w; \pi) := d_{\text{sem}}(\Phi(w\pi), \Phi(\pi)).$$

Proposition 8.5 (Explicit commutator residue)**.** *Define the commutator word*

$$c_{u,v} := \delta_v^{-1} \circ \gamma_u^{-1} \circ \delta_v \circ \gamma_u.$$

Then

$$c_{u,v}(\alpha, \beta, p) = (\alpha, \beta, p + uv).$$

Proof. Direct calculation:

$$(\alpha, \beta, p) \xrightarrow{\gamma_u} (\alpha + u, \beta, p) \xrightarrow{\delta_v} (\alpha + u, \beta, p + v(\alpha + u))$$
$$\xrightarrow{\gamma_u^{-1}} (\alpha, \beta, p + v(\alpha + u)) \xrightarrow{\delta_v^{-1}} (\alpha, \beta, p + uv).$$

□

Corollary 8.6 (Residue of the commutator word)**.** *For* $\pi = (\alpha, \beta, p) \in D_\eta$*,*

$$\mathrm{Res}_{\mathrm{obs}}(c_{u,v}; \pi) = (1 - \alpha - \beta)\,|uv| \geq \eta |uv|.$$

In particular, for $u, v \geq 0$*,*

$$\mathrm{Res}_{\mathrm{obs}}(c_{u,v}; \pi) = (1 - \alpha - \beta)\,uv \geq \eta uv.$$

Proof. The endpoint is $c_{u,v}\pi = (\alpha, \beta, p+uv)$. The first two semantic coordinates remain unchanged. The third changes by

$$\theta(\alpha, \beta, p + uv) - \theta(\alpha, \beta, p) = (1 - \alpha - \beta)uv.$$

The residue is the absolute value of this quantity. Use $1 - \alpha - \beta \geq \eta$. □

Definition 8.7 (Balanced-word admissibility set and radius)**.** Define

$$S_\pi := \left\{r > 0 : c_{r/2,r/2} \text{ is defined and all intermediate states of the word lie in } B_r^{\mathrm{prot}}(\pi)\right\},$$

and

$$r_{\mathrm{word}}(\pi) := \sup S_\pi.$$

Lemma 8.8 (Initial-interval property of the admissible set)**.** *For every* $\pi \in D_\eta$*, the set* S_π *is downward closed:*

$$r \in S_\pi,\ 0 < s < r \Longrightarrow s \in S_\pi.$$

Consequently, if $r_{\mathrm{word}}(\pi) > a$*, then there exists* $r \in S_\pi$ *with* $a < r < r_{\mathrm{word}}(\pi)$*.*

Proof. Let $\pi = (\alpha, \beta, p)$. The intermediate states of $c_{u,v}$ are

$$(\alpha + u, \beta, p), \qquad (\alpha + u, \beta, p + v(\alpha + u)), \qquad (\alpha, \beta, p + v(\alpha + u)),$$

and the endpoint is

$$(\alpha, \beta, p + uv).$$

For the balanced word $u = v = t/2$, each coordinate increment is a nondecreasing function of $t \geq 0$:

$$u = \frac{t}{2}, \qquad v(\alpha + u) = \frac{t}{2}\left(\alpha + \frac{t}{2}\right), \qquad uv = \frac{t^2}{4}.$$

Hence, if the admissibility inequalities defining D_η and the ball-inclusion inequalities defining $B_t^{\mathrm{prot}}(\pi)$ hold at $t = r$, then they also hold at every $0 < s < r$, because the corresponding increments at s are bounded above by those at r. Thus S_π is downward closed.

If $r_{\mathrm{word}}(\pi) > a$, then by definition of supremum there exists $r \in S_\pi$ with $a < r \leq r_{\mathrm{word}}(\pi)$. Since S_π is downward closed, any $r' \in (a, r)$ also lies in S_π. □

Theorem 8.9 (Finite upper agreement bound in the binary model)**.** *If*

$$r_{\mathrm{word}}(\pi) > \sqrt{\frac{4\varepsilon}{\eta}},$$

then

$$R_\varepsilon(\pi) \leq \sqrt{\frac{4\varepsilon}{\eta}}.$$

Proof. Assume by contradiction that

$$R_\varepsilon(\pi) > \sqrt{\frac{4\varepsilon}{\eta}}.$$

By the previous lemma, choose

$$r \in S_\pi \quad \text{such that} \quad \sqrt{\frac{4\varepsilon}{\eta}} < r < \min\{R_\varepsilon(\pi), r_{\text{word}}(\pi)\}.$$

Since $r < R_\varepsilon(\pi)$, every state in $B_r^{\text{prot}}(\pi)$ is ε-close semantically to π. Because $r \in S_\pi$, all intermediate states and the endpoint of the balanced commutator word $c_{r/2,r/2}$ lie in the ball. Therefore

$$\text{Res}_{\text{obs}}(c_{r/2,r/2}; \pi) \leq \varepsilon.$$

But the previous corollary yields

$$\text{Res}_{\text{obs}}(c_{r/2,r/2}; \pi) \geq \eta \frac{r^2}{4} > \varepsilon,$$

contradiction. ☐

Corollary 8.10 (Two-sided agreement window)**.** *If*

$$r_{\text{word}}(\pi) > \sqrt{\frac{4\varepsilon}{\eta}},$$

then

$$\min\left(r_{\text{adm}}(\pi), \frac{\varepsilon}{2}\right) \leq R_\varepsilon(\pi) \leq \sqrt{\frac{4\varepsilon}{\eta}}.$$

Proof. Combine the lower and upper bounds. ☐

9 Order induced by non-closure

Definition 9.1 (Obstruction vector)**.** Let Z be a class of semantic states. For each resource level r, fix five nonnegative obstruction functionals

$$\kappa_r, \qquad \text{Hol}_r^{\text{atlas}}, \qquad \text{Hol}_r^{\text{prot}}, \qquad \mathfrak{g}_r, \qquad \mathfrak{c}_r$$

on Z. Define

$$\Delta_r(z) = \big(\kappa_r(z),\ \text{Hol}_r^{\text{atlas}}(z),\ \text{Hol}_r^{\text{prot}}(z),\ \mathfrak{g}_r(z),\ \mathfrak{c}_r(z)\big) \in [0, \infty)^5.$$

Definition 9.2 (Refinement preorder)**.** For $x, y \in Z$, define

$$x \preceq_r y \iff \Delta_r(y) \leq \Delta_r(x) \quad \text{componentwise.}$$

Define

$$x \sim_r y \iff x \preceq_r y \text{ and } y \preceq_r x.$$

Proposition 9.3. *The relation $\preceq_r$ is a preorder, and*

$$x \sim_r y \iff \Delta_r(x) = \Delta_r(y).$$

Proof. Immediate from the componentwise order on $[0, \infty)^5$. $\square$

Definition 9.4 (Global trivialization)**.** A class Z is globally trivialized at scale r if there exists a global chart, section, or normalization on Z such that

$$\Delta_r(z) = 0 \qquad \forall z \in Z.$$

Proposition 9.5 (Global trivialization collapses the quotient)**.** *If Z is globally trivialized at scale r, then $Z/{\sim_r}$ has a single class.*

Proof. If $\Delta_r(z) = 0$ for all z, then $\Delta_r(x) = \Delta_r(y)$ for all x, y, hence $x \sim_r y$. $\square$

Theorem 9.6 (Persistent obstruction prevents global trivialization)**.** *If*

$$\inf_{z \in Z} \|\Delta_r(z)\|_\infty > 0,$$

then Z is not globally trivialized at scale r.

Proof. Global trivialization would imply $\Delta_r(z) = 0$ for all z. $\square$

Theorem 9.7 (Non-constant obstruction prevents collapse)**.** *If Δ_r is not constant on Z, then $Z/{\sim_r}$ has at least two distinct classes.*

Proof. If Δ_r is not constant, choose $x, y \in Z$ with $\Delta_r(x) \neq \Delta_r(y)$. Then $x \not\sim_r y$. $\square$

Definition 9.8 (Deployment path)**.** A deployment path at scale r is a chain

$$z_0 \preceq_r z_1 \preceq_r \cdots \preceq_r z_n.$$

10 Conditional Yang–Mills reduction

10.1 Spectral implication of exponential clustering

Let $(\mathcal{H}_{\text{phys}}, \Omega, H)$ be a Euclidean-OS reconstructed gauge-invariant physical Hilbert space. For a local gauge-invariant observable O, define

$$\widetilde{O} := O - \langle \Omega, O\Omega \rangle,$$

and

$$C_O(t) := \langle \Omega, \widetilde{O}\, e^{-tH}\, \widetilde{O}\, \Omega \rangle.$$

Definition 10.1 (Spectrally separating family)**.** A family $\mathcal{M}$ of local gauge-invariant observables is *spectrally separating* if

$$\overline{\operatorname{span}\{\widetilde{O}\,\Omega : \ O \in \mathcal{M}\}} = \mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega.$$

Proposition 10.2 (Exponential clustering implies spectral mass gap)**.** *Suppose there exists $c > 0$ such that for a spectrally separating family $\mathcal{M}$ of local gauge-invariant observables,*

$$C_O(t) \le A_O e^{-ct} \qquad \forall O \in \mathcal{M}, \ \textit{for } t \gg 1.$$

Then

$$\operatorname{Spec}(H) \subseteq \{0\} \cup [c, \infty),$$

hence the theory has mass gap $\Delta \ge c$.

Proof. By the spectral theorem,

$$C_O(t) = \int_{[0,\infty)} e^{-tm}\, d\mu_O(m)$$

for a positive measure μ_O. If $\mu_O((0,c)) > 0$, then there exists $\delta > 0$ with

$$\mu_O([0, c-\delta]) > 0.$$

Hence

$$C_O(t) \geq \mu_O([0, c-\delta])e^{-(c-\delta)t},$$

contradicting the assumed upper bound $A_O e^{-ct}$ for large t. Therefore $\mu_O((0,c)) = 0$ for all $O \in \mathcal{M}$.

Let $E_{(0,c)}$ denote the spectral projector of H on $(0,c)$. Suppose for contradiction that

$$E_{(0,c)} \neq 0 \quad \text{on} \quad \mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega.$$

Then there exists a nonzero vector in $\mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega$ with nontrivial projection onto $(0,c)$. Since $\mathcal{M}$ is spectrally separating,

$$\overline{\operatorname{span}\{\widetilde{O}\,\Omega : \ O \in \mathcal{M}\}} = \mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega,$$

so there must exist some $O \in \mathcal{M}$ such that

$$E_{(0,c)}\, \widetilde{O}\,\Omega \neq 0.$$

For this observable,

$$\mu_O((0,c)) = \langle \widetilde{O}\Omega, E_{(0,c)}\widetilde{O}\Omega \rangle > 0,$$

contradicting the previous paragraph. Hence

$$E_{(0,c)} = 0 \quad \text{on} \quad \mathcal{H}_{\text{phys}} \ominus \mathbb{R}\Omega,$$

which is equivalent to

$$\operatorname{Spec}(H) \subseteq \{0\} \cup [c, \infty).$$

Therefore the theory has mass gap $\Delta \geq c$. □

10.2 Audited Yang–Mills interface

Definition 10.3 (Certificate class)**.** Let Σ^{cert} be a class of admissible certificates, models, or protocol representations together with a cost map

$$\text{cost} : \Sigma^{\text{cert}} \to [0, \infty).$$

Definition 10.4 (Audited Yang–Mills interface)**.** An audited Yang–Mills interface is a tuple

$$S_{\text{YM}} = (\Theta, \Sigma_r, \Sigma^{\text{cert}}, \mathcal{M}_r, \text{err}, \mu, r),$$

where:

1. Θ is a parameter space (a, L, β, m_q, ξ) of lattice spacing, volume, bare couplings, masses, and protocol data;

2. Σ_r is a finite bank of lattice gauge configurations and admissible protocol operations at resource level r;

3. Σ^{cert} is the admissible certificate class;

4. $\mathcal{M}_r$ is a class of gauge-invariant observables;

5. err is a certification error functional

$$\mathrm{err} : \Sigma^{\mathrm{cert}} \times \mathcal{C}_{\mathrm{IR}}^{\mathrm{g.i.}} \to [0, \infty);$$

6. μ is an admissibility measure.

Definition 10.5 (Infrared curvature floor)**.** For the infrared gauge-invariant semantic class $\mathcal{C}_{\mathrm{IR}}^{\mathrm{g.i.}}$, define

$$\kappa_r^{\sup}(\mathcal{C}_{\mathrm{IR}}^{\mathrm{g.i.}}) := \sup_{o \in \mathcal{C}_{\mathrm{IR}}^{\mathrm{g.i.}}} \inf_{\sigma \in \Sigma^{\mathrm{cert}} :\ \mathrm{cost}(\sigma) \leq r} \mathrm{err}(\sigma, o),$$

and

$$\kappa_*^{\sup}(\mathcal{C}_{\mathrm{IR}}^{\mathrm{g.i.}}) := \liminf_{r \to \infty} \kappa_r^{\sup}(\mathcal{C}_{\mathrm{IR}}^{\mathrm{g.i.}}).$$

Definition 10.6 (UV/IR protocol subcategory)**.** Let

$$\mathbf{Prot}_{UV/IR} \subseteq \mathbf{Prot}$$

be the smallest subcategory generated by five specified families of morphisms:

1. continuum extrapolation morphisms;

2. volume extrapolation morphisms;

3. renormalization morphisms;

4. observable-choice morphisms;

5. finite-bank transfer morphisms.

Definition 10.7 (Nontrivial UV/IR abstract holonomy)**.** We say that $\mathbf{Prot}_{UV/IR}$ has *nontrivial UV/IR abstract holonomy* if there exist a protocol state $\pi \in \mathrm{Ob}(\mathbf{Prot}_{UV/IR})$ and a loop ℓ at π in $\mathbf{Prot}_{UV/IR}$ such that

$$\mathrm{Hol}_{\mathrm{abs}}(\ell; \pi) > 0.$$

Hypothesis 10.8 (Finite-bank transfer to the ideal continuum)**.** There exists a transfer modulus $\omega_r \to 0$ carrying finite-bank protocol semantics to the ideal continuum regime.

Conjecture 10.9 (Bridge A: abstract holonomy to curvature)**.** Persistence of nontrivial UV/IR abstract holonomy implies

$$\mathbf{Prot}_{UV/IR} \text{ has nontrivial UV/IR abstract holonomy} \Longrightarrow \kappa_*^{\sup}(\mathcal{C}_{\mathrm{IR}}^{\mathrm{g.i.}}) > 0.$$

Conjecture 10.10 (Bridge B: curvature to exponential clustering)**.** A positive infrared curvature floor implies the existence of $c > 0$ and a spectrally separating family $\mathcal{M}$ of local gauge-invariant observables such that

$$C_O(t) \leq A_O e^{-ct} \qquad \forall O \in \mathcal{M}.$$

Theorem 10.11 (Conditional Yang–Mills reduction theorem)**.** *Assume:*

1. *a reflection-positive audited lattice family and OS reconstruction;*

2. *finite-bank transfer to the ideal continuum;*
3. *persistence of nontrivial UV/IR abstract holonomy;*
4. *Bridge A and Bridge B.*

Then the reconstructed Yang–Mills theory has a strictly positive mass gap.

Proof. By persistence of nontrivial UV/IR abstract holonomy and Bridge A,

$$\kappa_*^{\sup}(\mathcal{C}_{\mathrm{IR}}^{\mathrm{g.i.}}) > 0.$$

Bridge B then yields exponential clustering for a spectrally separating family of gauge-invariant observables. By the preceding proposition,

$$\mathrm{Spec}(H) \subseteq \{0\} \cup [c, \infty),$$

so the mass gap satisfies $\Delta \geq c > 0$. □

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